AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

ANNUAL EXAMINATIONS 2021

Biology

Total Time: 1 hour 40 minutes

Total Marks: 50 (40-Theory & 10-Alternate to Practical)

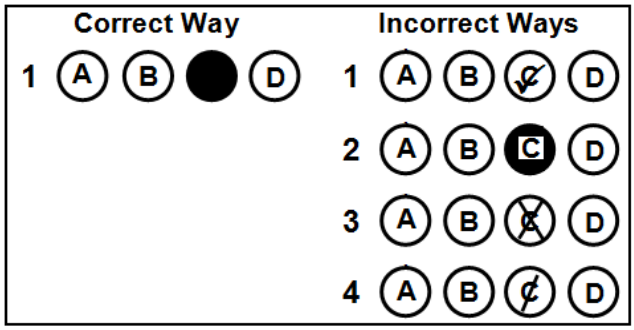
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
40 MCQs	10 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 40 MCQs will be equated to the total marks of 65 for the theory examination results.
9. You may use a simple calculator if you wish.

THEORY (Questions 1-40)

1. The phase of meiotic division in which homologous chromosomes start to move towards the opposite poles is

   A. anaphase I.
   B. telophase I.
   C. anaphase II.
   D. telophase II.

2. How many daughter cells are produced as the result of meiosis-I?

   A. Two daughter cells with similar genetic makeup
   B. Two daughter cells with distinct genetic makeup
   C. Four daughter cells with similar genetic makeup
   D. Four daughter cells with distinct genetic makeup

3. A biologist treated a cell with a certain type of chemical that inhibited the cell's ability to synthesise DNA.

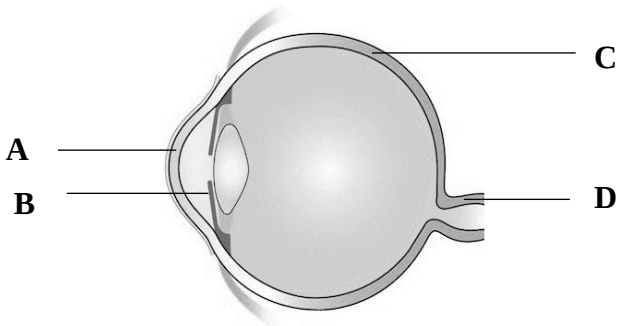
   If this treated cell undergoes the process of cell cycle, then the affected phase would be

   A. S phase.
   B. G0 phase.
   C. G1 phase.
   D. G2 phase.

4. The part of neuron which contains the major portion of cytoplasm is

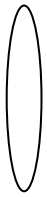
   A. axon.
   B. dendrites.
   C. cell body.
   D. myelin sheath.

5. In the given vertical section diagram of the human eye, the part which causes refraction of light is labelled as



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6. Diagrams **I** and **II** represent the conditions of pupil while **III** and **IV** show the shape of eye lens.

			
I	**II**	**III**	**IV**

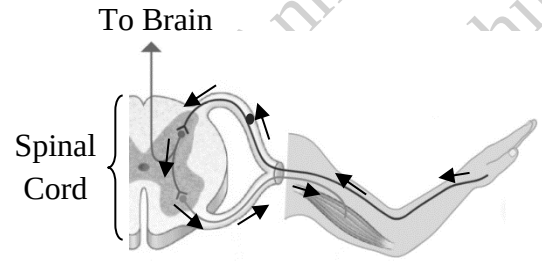
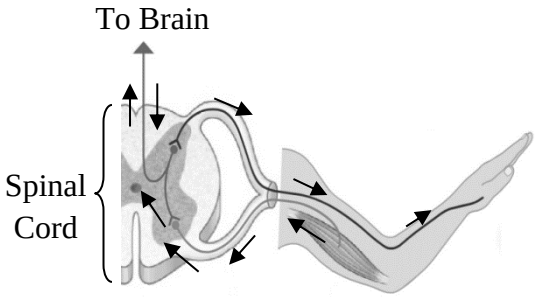
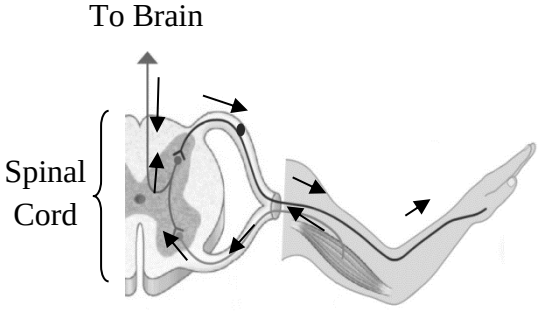
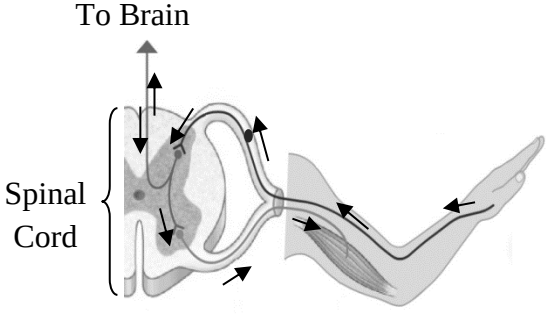
The diagrams which show the condition of pupil and shape of eye lens when looking at the moon are

A. I and III.
B. I and IV.
C. II and III.
D. II and IV.

7. In human beings, deficiency of vitamin A causes night blindness. This is because vitamin A is required for the production of

A. iodopsin.
B. rhodopsin.
C. vitreous humour.
D. aqueous humour.

8. What will be the pathway for a spinal reflex action when the hand of an individual is pricked by a needle?

	
A	B
	
C	D

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9. The gland which is located in the human brain is

   A. adrenal.
   B. thyroid.
   C. pituitary.
   D. parathyroid.

10. In the human body, the function of progesterone is to increase the

   A. heart rate during exercise.
   B. breathing rate during emergency situation.
   C. thickness of uterine wall during pregnancy.
   D. production of gametes during reproductive cycle.

11. The given symptoms are observed in a patient with over-production of a hormone.

   - Shivering of hands
   - Frequent sweating
   - Increase in heartbeat
   - Increase in energy production

   The hormone identified is

   A. insulin.
   B. thyroxin.
   C. glucagon.
   D. calcitonin.

12. After consuming a carbohydrate rich meal by a non-diabetic person, the blood level of glucagon and insulin will

	Glucagon Level	**Insulin Level**
A	decrease	increase
B	increase	decrease
C	decrease	decrease
D	increase	increase

13. When Aleem encounters a fighter bull, he experiences a sudden increase in his heart beat. This increase in the heart beat would

   A. increase the rate of digestion.
   B. decrease the air flow to his lungs.
   C. convert maximum glucose to glycogen.
   D. increase the supply of oxygen to his muscles.

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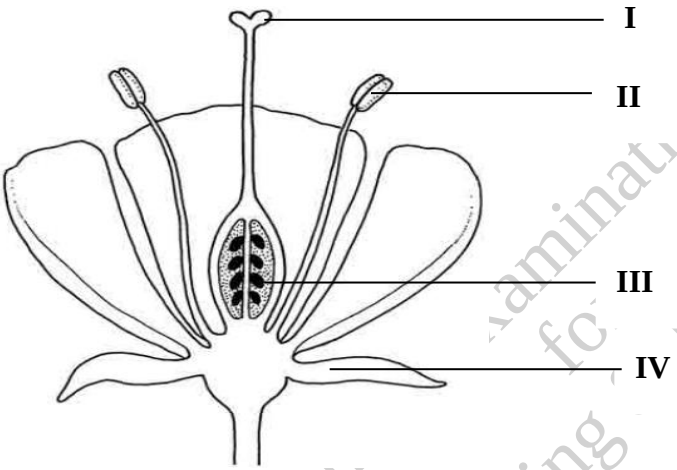
14. The drones of honey bee families are haploid because they are produced through the process of

A. budding.
B. binary fission.
C. multiple fission.
D. parthenogenesis.

15. In a flower, the site where germination of pollen grains takes place is

A. style.
B. ovule.
C. anther.
D. stigma.

16. In pollination, pollens transfer from the labelled parts



A. II to I.
B. I to III.
C. II to IV.
D. III to IV.

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17. Given are the characteristics of a type of asexual reproduction.

I. Parent cells are surrounded by cyst.
II. It occurs mostly in unfavourable conditions.
III. Nucleus and cytoplasm divide simultaneously.

The characteristic(s) of multiple fission is/ are

A. I only.
B. III only.
C. I and II.
D. II and III.

18. In contrast to spermatogenesis, oogenesis

A. produces one mature gamete.
B. involves mitosis and meiosis.
C. involves the maturation of germ cells.
D. completes the meiotic division before fertilisation.

19. In a healthy male rabbit, the total number of sperms that are expected to be produced by 100 primary spermatocytes are

    A. 100
    B. 200
    C. 400
    D. 500

20. In the human body, the set of structures which is composed of cartilage is

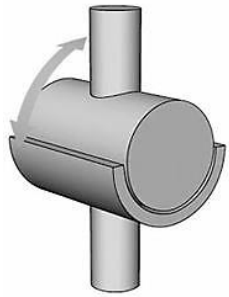
    A. epiglottis, trachea and nose tip.
    B. ear pinnae, larynx and pharynx.
    C. pharynx, larynx and oesophagus.
    D. epiglottis, oesophagus and trachea.

21. Suppose, an athlete is diagnosed with a fracture in the sternum.

    The part of skeleton that would be affected in this case is the

    A. skull.
    B. rib cage.
    C. pelvic girdle.
    D. pectoral girdle.

22. The given diagram represents the working of one of the types of joints present in the human skeleton.

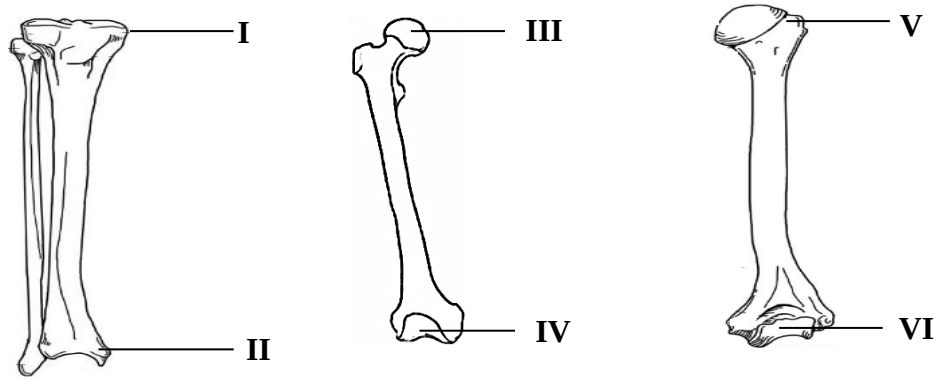


    The pair of joints that shows this type of working is

    A. hip and knee.
    B. elbow and knee.
    C. shoulder and hip.
    D. shoulder and elbow.

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23. The given diagram shows the anterior view of different bones of the human skeleton



The parts which form the knee joint are labelled as

A. IV and I.
B. III and II.
C. V and IV.
D. III and VI.

24. The MRI (magnetic resonance imaging) diagnostic report of a joint reveals the destruction of a cartilage.

This condition is diagnosed as

A. fracture.
B. dislocation.
C. osteoporosis.
D. osteoarthritis.

25. If a DNA segment contains 30 guanine and 60 thymine in a single strand, then the total number of nucleotides in this DNA segment will be

A. 90
B. 120
C. 150
D. 180

26. If the nucleotide order in DNA segment is GCCATA, then the total number of hydrogen bonds in this segment will be

A. 6
B. 12
C. 15
D. 18

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27. If an individual homozygous for a trait is crossed with an individual homozygous recessive for the same trait, then the phenotype of F1 generation will be

A. 100% heterozygous dominant.
B. 75% heterozygous dominant.
C. 50% homozygous recessive.
D. 25% homozygous recessive.

28. Red flower (R) colour is completely dominant over white flower (r) colour in a particular plant.

When a heterozygous red flower is crossed with a white flower, the probable percentage for the offspring flower colour would be

A. 100% red.
B. 75% red and 25% white.
C. 50% red and 50% white.
D. 25% red and 75% white.

29. The given table shows the result obtained in an experiment of breeding cattle with unknown genotype for their coat colour.

(**Note**: The allele for black hair coat colour is completely dominant over the allele for red hair coat colour and these are represented as **B** and **b**, respectively.)

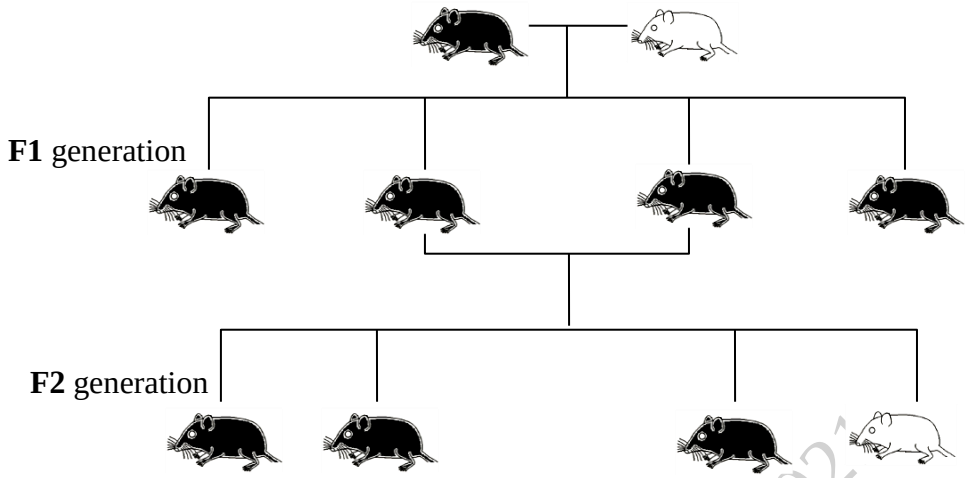
Cross	**Parents**		**Offspring**	
	Cow	**Bull**	**% of Black Hair Coat Colour**	**% of Red Hair Coat Colour**
I	Black	Red	50	50
II	Black	Black	75	25
III	Black	Red	100	0

If the parental black cow from cross **III** is bred with the parental black bull from cross **II**, then which of the following would be TRUE for the genotypes of their offspring?

	Heterozygous Offspring	**Homozygous Offspring**
A	0	100
B	25	75
C	50	50
D	75	25

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30. The pedigree shows the inheritance of fur colour in pure breeding rats.



In the given pedigree, the total number of rats that are homozygous are

A. 2
B. 4
C. 6
D. 8

31. The option that CORRECTLY identifies the inheritance of the co-dominant trait from the parents to the offspring is

	Parent Phenotype	Offspring's Phenotype
A	straight and curly hair texture	wavy hair
B	red and white Japanese 4 o' clock plants	pink colour
C	AB and A blood group	AB blood group
D	free earlobe and attached earlobe	free earlobe

32. If a mother has AB blood group and the father has B blood group, then the type of blood group that the children will NEVER have is

A. O.
B. A.
C. B.
D. AB.

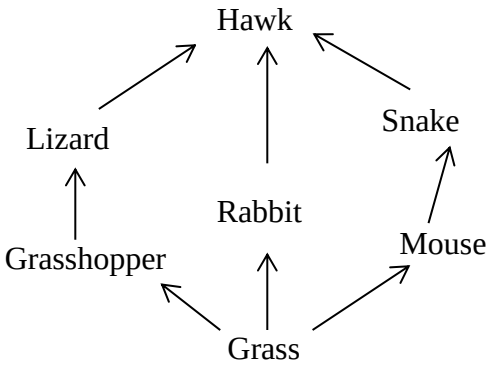
33. In an experiment, healthy mice were exposed to some radiations. When they were allowed to mate, they produced offspring without tails.

The reason for the lack of tail would be

A. mutation.
B. evolution.
C. continuous variation.
D. discontinuous variation.

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34. Consider the given food web.



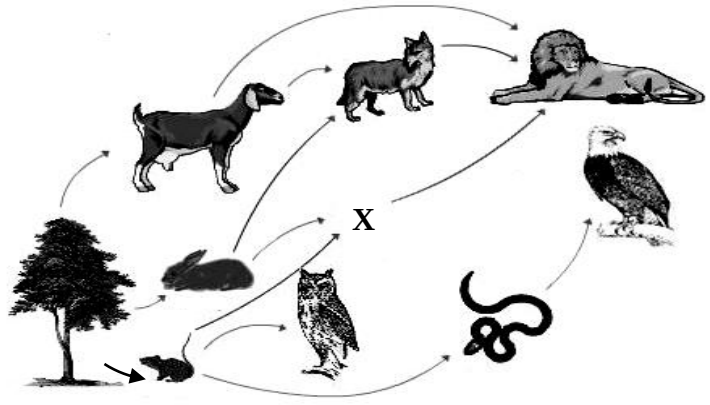
Which of the following combinations is CORRECT?

	Organism	**Trophic Level**
A	Hawk	Secondary consumer
B	Snake	Primary consumer
C	Grass	Primary consumer
D	Mouse	Secondary consumer

35. The function of fungi in an ecosystem is to

A. provide oxygen gas to producers.
B. return nutrients to the environment.
C. increase the complexity of food chains.
D. decrease the competition among consumers.

36. In the given food web, the living organism **X** could be a/ an



A. cat.
B. cow.
C. zebra.
D. elephant.

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37. What happens in commensalism?

A. Both species are benefited.
B. Both species compete for limited resources.
C. One species benefits while the other remains unaffected.
D. One species benefits while the other is negatively affected.

38. The process which makes the use of carbon dioxide present in air is the

A. burning of fossil fuels on land.
B. production of glucose in plants.
C. assimilation of nutrients in animals.
D. breakdown of organic matter by decomposers.

39. The stage of nitrogen cycle which converts atmospheric nitrogen into useable form for plants is

A. nitrification.
B. denitrification.
C. ammonification.
D. nitrogen fixation.

40. In the environmental management, the given logos represent



Logo **I**



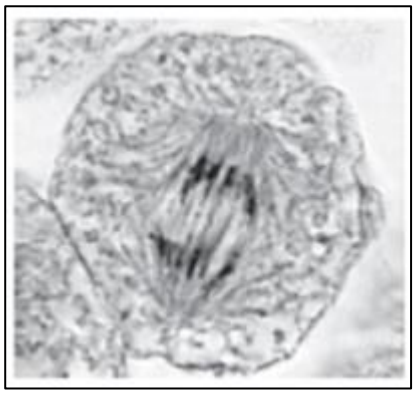
Logo **II**

	Logo I	**Logo II**
A	reuse	recycle
B	recycle	reduce
C	reduce	reuse
D	recycle	reuse

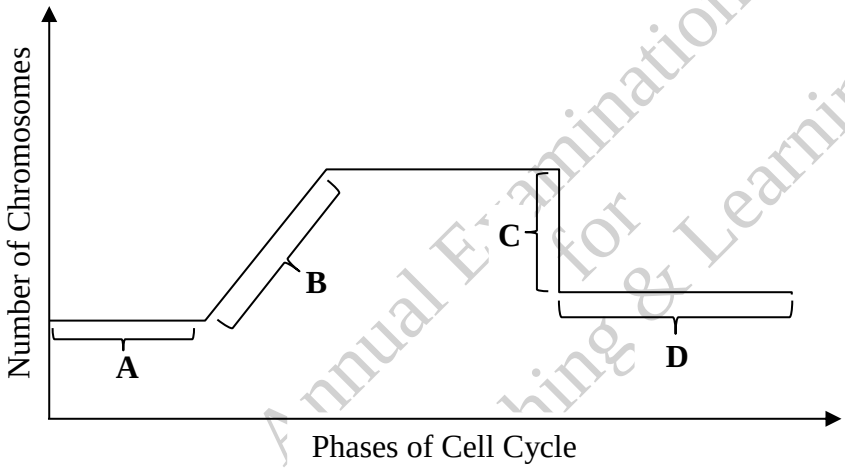
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ALTERNATE TO PRACTICAL (ATP: Questions 41-50)

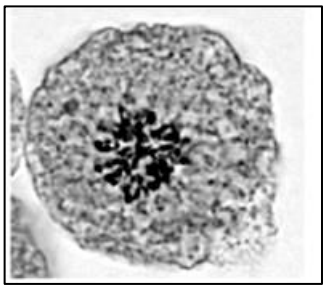
41. The given photomicrograph depicts one of the phases of the cell cycle.



This phase takes place at the labelled part



42. The given photomicrograph depicts one of the phases of mitosis.



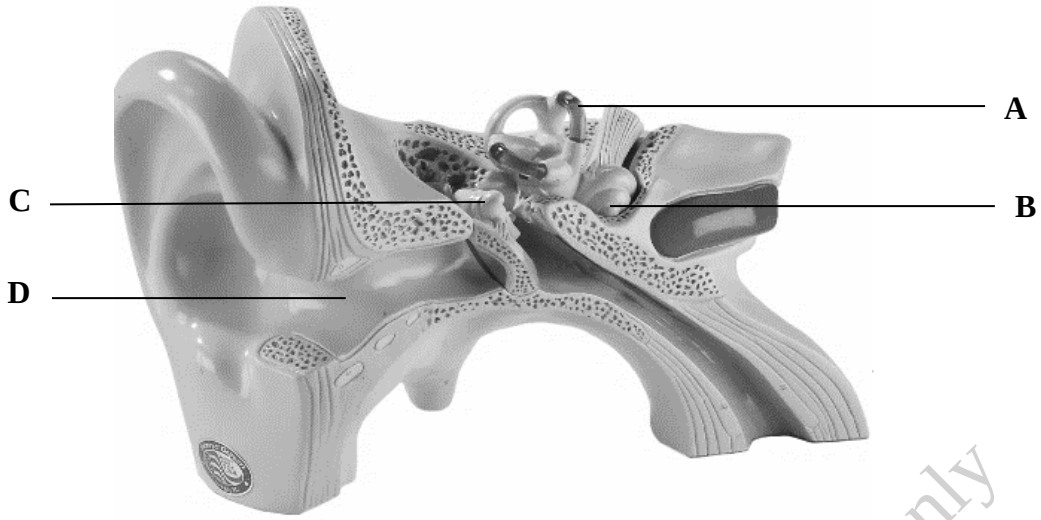
If the chromosome number is **X** in this phase of mitosis, then the number of chromosomes in the daughter nuclei after the telophase would be

A. X
B. 2X
C. X/4
D. X/2

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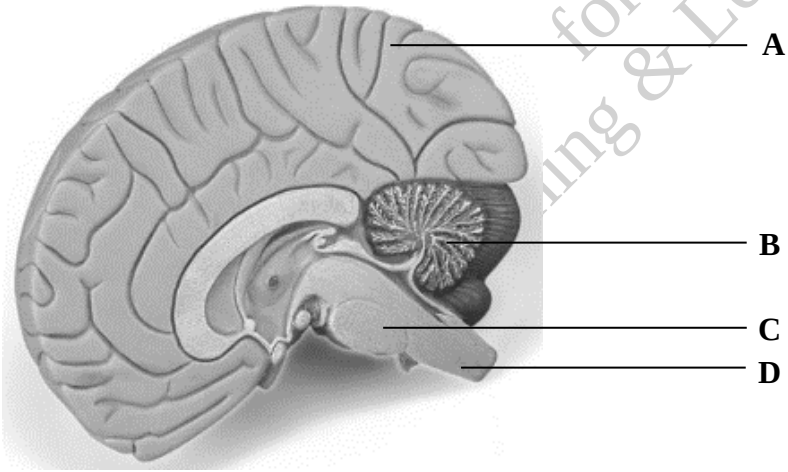
43. The given image represents a model of the human ear.

Which of the labelled part would be affected if a patient has an infection in the middle ear?



44. After the brain injury in an accident, a person is unable to maintain and coordinate motor functions.

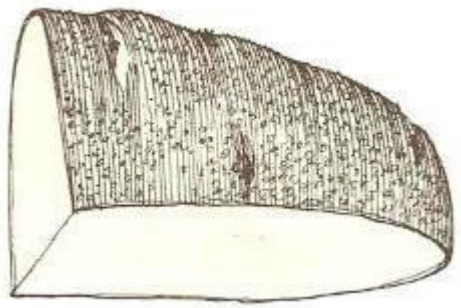
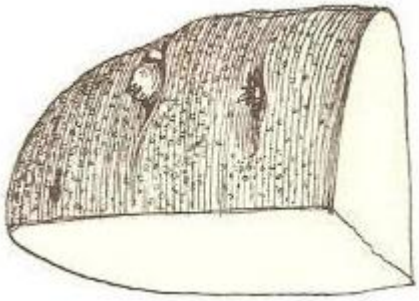
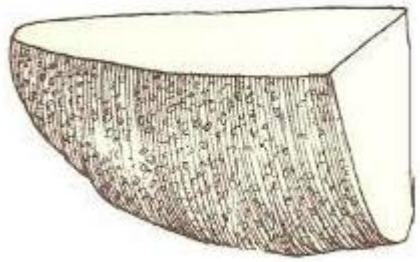
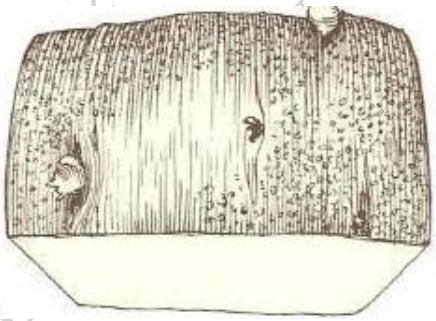
The part of the brain that has been affected in this case is labelled as



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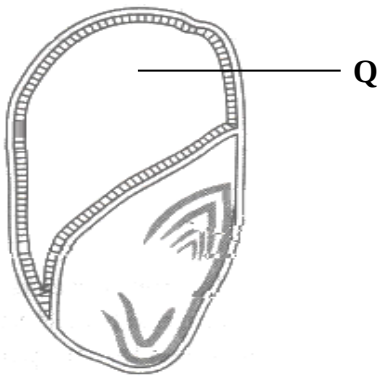
45. The given images represent different pieces of potato tubers.

The piece that which would NOT be able to produce a stem is

	
A	B
	
C	D

46. The given diagram represents longitudinal section of a maize seed.

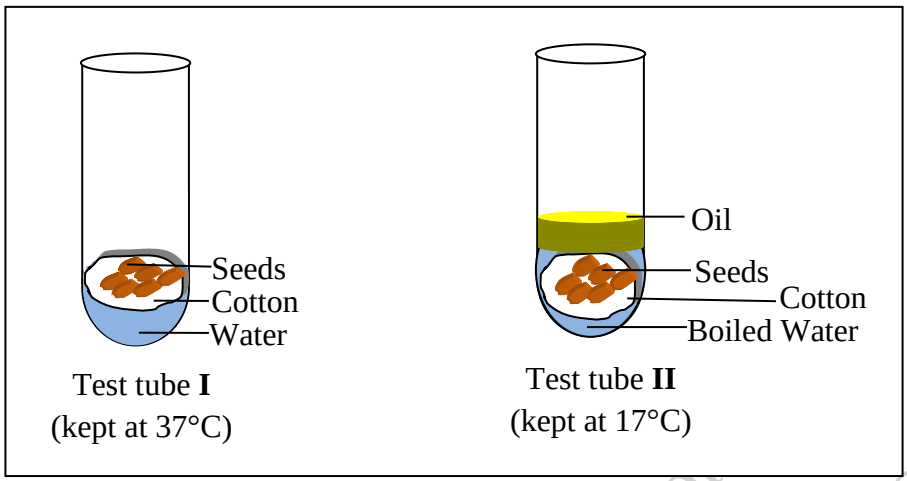
The labelled part **Q** is



A. coleoptile.
B. epithelium.
C. scutellum.
D. endosperm.

47. The given diagram shows an experiment to study the conditions necessary for the germination of seeds.

The seeds in test tube **I** would germinate while there would be no germination in test tube **II**.

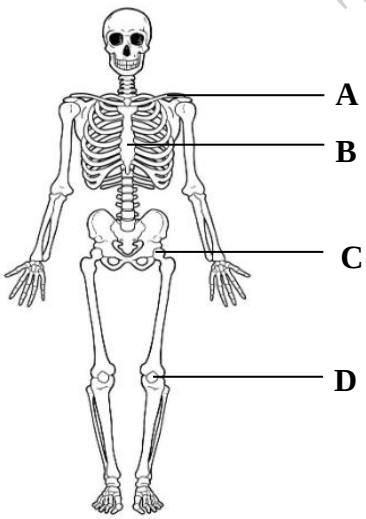


In the test tube **II**, the seeds will not germinate because

I. oxygen is absent in water
II. 17°C temperature is inappropriate for seed germination
III. cotton absorbed the required amount of water for germination

A. I only.
B. III only.
C. I and II.
D. II and III.

48. In the given image of the human skeleton, patella is labelled as

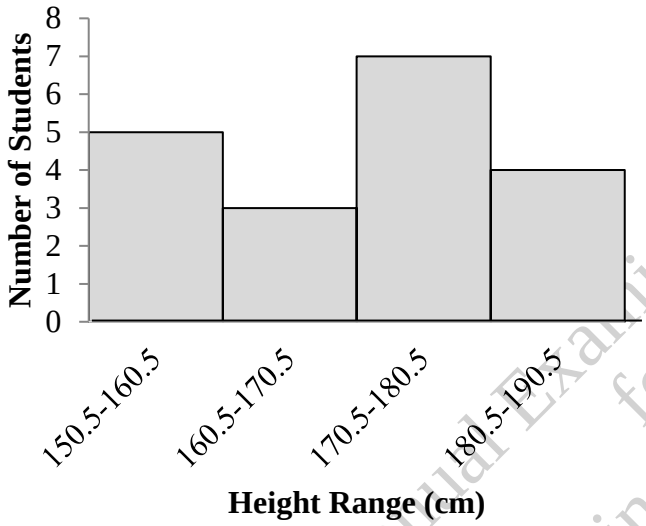


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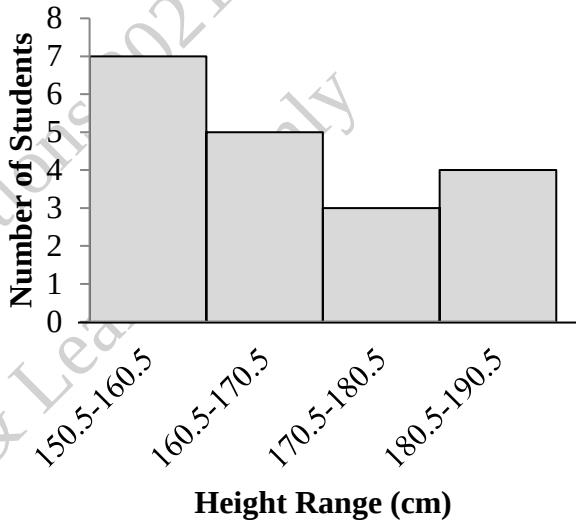
49. Sana obtained the following data related to the height (cm) of her class fellows.

Height Range (cm)	Number of Students
150.5-160.5	5
160.5-170.5	3
170.5-180.5	7
180.5-190.5	4

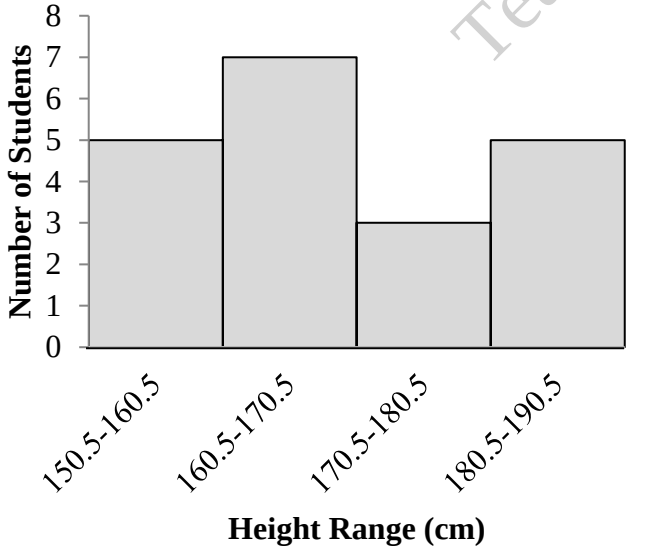
The CORRECT graphical representation of this data is



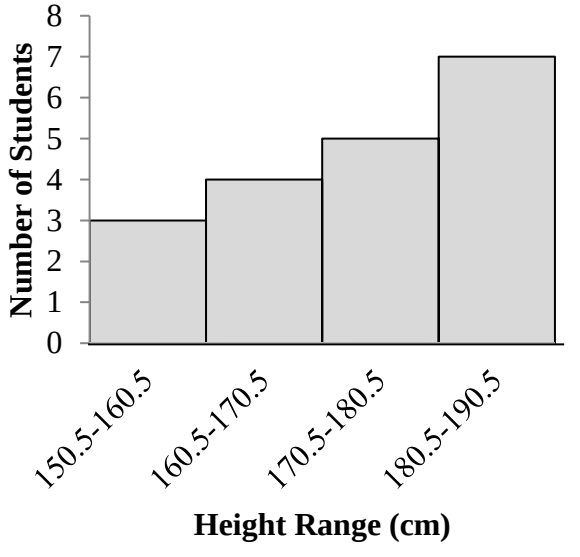
A



B



C



D

50. The given data represents the height (cm) of grade 10 students.

100.5 106.6 110.9 119.8 125.6 130.4 130.6 134.5 138.9 140 145.9

147.8 150.4 155.5 157.9 160.1 165.3 169.2 170.6 180.4 190.8 200.5

205.5 210.6 215.9 220.6

The height range which contains the HIGHEST number of students is

A. 100.5-130.5
B. 130.5-160.5
C. 160.5-190.5
D. 190.5-220.5

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